

A Reproducible Binary-Schedule Benchmark for Robust Low-Thrust Cislunar Initialization Under Missed-Thrust Events

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Abstract

This paper defines a reproducible normalized Earth–Moon CR3BP missed-thrust benchmark for robust low-thrust cislunar initialization. It is a preliminary-design algorithm benchmark, not high-fidelity validation or flight targeting: the configured terminal-error thresholds are normalized screening tolerances. Under equal true trajectory-evaluation budgets, the benchmark compares classical direct-objective initializers, surrogate-QUBO sampling, simulated statevector QAOA, and continuous recovery diagnostics. The primary negative result is that the evidence does not support quantum or QAOA superiority: the configured QAOA initializer is weaker than the strongest classical initializers in the 30-seed main-method package, and optimized simulated QAOA is not statistically superior to surrogate-QUBO simulated annealing in the focused ablation. The primary positive result is narrower and useful for trajectory-optimization researchers: a duty-cycle prior steers several samplers toward refinable high-duty schedules, while a scoped locked-nominal tail-coast continuous backend meets the configured one- and two-segment mask screening thresholds. That tail-coast result uses a final-five-segment coast, deterministic fallback starts, and two late-tail direct evaluations without recovery variables; its persisted accepted controls also replay deterministically under normalized CR3BP with zero terminal-error delta across the nominal row plus 27 branch rows. This remains continuous-backend and accepted-control replay evidence only, not fuel optimality, high-fidelity robustness, production solver parity, or quantum evidence.

Keywords: Trajectory optimization benchmark, low-thrust trajectory optimization, cislunar dynamics, missed-thrust recovery, robust

1. Introduction

Electric low-thrust propulsion enables propellant-efficient cislunar transfers, but it also makes trajectory design highly sensitive to the initial guess supplied to a nonlinear optimizer. A practical design must determine both continuous thrust magnitudes and directions and a qualitative maneuver structure: which intervals thrust, coast, or recover after an interruption. For astrodynamics researchers, this creates a reproducibility problem as much as an algorithm problem. A proposed initializer is difficult to interpret unless the schedule objective, missed-thrust masks, continuous recovery backend, evaluation budget, and failure cases are all visible.

Missed-thrust robustness is a demanding test of that pipeline. Safe modes, propulsion interruptions, or operations-driven thrust outages can invalidate a nominal low-thrust transfer. Grebow and McCarty show that including missed-thrust branches in low-thrust cislunar design can greatly reduce recovery propellant for week-scale outages [1]. Sinha and Beeson identify initial-guess generation as a bottleneck for low-thrust trajectory design with missed-thrust robustness [2]. This paper therefore asks a benchmark question rather than a quantum-performance question:

Can a reproducible binary-schedule benchmark reveal which missed-thrust initializers produce refinable schedules, which failure modes are caused by the continuous backend, and how surrogate-QUBO and simulated-QAOA initializers compare with strong classical samplers under equal trajectory-evaluation budgets?

Binary and QUBO formulations are included because they are a natural way to encode thrust-window availability, and because recent low-thrust studies have explored binary/QUBO and quantum-assisted formulations [3, 4]. They are not the headline claim. In this paper QUBO and QAOA are evaluated initializer families inside a broader robust-trajectory benchmark; the continuous trajectory remains a classical optimal-control problem, and negative evidence is reported as part of the contribution.

The paper makes four contributions for a trajectory-optimization audience. First, it defines and archives a normalized Earth–Moon CR3BP missed-

thrust binary-schedule benchmark with source states, configurations, generated tables, figures, result metadata, and an artifact manifest. Second, it reports an equal-budget 30-seed comparison of classical direct-objective initializers, surrogate-QUBO simulated annealing, simulated statevector QAOA, all-windows continuous refinement, and a post hoc threshold-sensitivity sanity check derived from the same raw 30-seed CSV. Third, it adds deterministic evidence-synthesis and recorded-control replay layers that separate initializer evidence from continuous-backend evidence, led by a scoped fixed-final-time tail-coast hard-catalog diagnostic that evaluates all configured one- and two-segment masks within a locked-nominal setup and replays the persisted accepted controls. Fourth, it records failure modes and claim boundaries: sparse schedules can be unrefinable, duty-cycle priors can recover high-duty schedule classes, all-windows continuous refinement can dominate selected-worst error, and the tested QAOA variants do not establish superiority over classical samplers.

The contribution is the schedule-initializer benchmark layer, not a replacement for production sequential-convex, collocation, or indirect low-thrust tools. Those mature solvers are downstream consumers and related-work comparators for a schedule or continuous-control initial guess. The controls used here—all-windows continuous refinement, direct shooting, compact collocation, continuation, and tail-coast diagnostics—are included to isolate whether the discrete schedule initializer adds value before handing the problem to stronger continuous machinery.

The claims are intentionally narrow. The experiments use normalized CR3BP dynamics, screening thresholds, configured outage families, and research-grade continuous backends. They do not establish quantum advantage, high-fidelity mission validation, fuel-optimal recovery, certified flight design, or universal missed-thrust robustness.

2. Related Work

2.1. Low-thrust trajectory optimization

Low-thrust trajectory optimization has a mature literature spanning indirect methods, direct transcription, collocation, global search, sequential convex programming, and hybrid approaches. Morante et al. survey low-thrust optimization as a hybrid optimal-control problem with both continuous and discrete structure [5]. In the Earth–Moon system, Ozimek and Howell computed low-thrust transfers to libration-point orbits and showed

the role of CR3BP structure in preliminary design [6]. Pritchett, Howell, and Grebow used collocation, trajectory stacking, orbit chaining, and continuation to compute low-thrust transfers in the restricted three-body problem, including NRHO-to-DRO examples [7]. More recent cislunar and deep-space solvers include successive convex optimization for libration-point transfers [8], convex-programming approaches for rapid low-thrust trajectory design [9], thrust-regularized convex optimization [10], indirect forward-backward shooting in complex dynamics [11], and adaptive-mesh sequential convex programming [12]. These methods are mature trajectory-design baselines; the present work is not proposed as a replacement for them.

For preliminary cislunar design, the Circular Restricted Three-Body Problem is widely used because it captures the nonlinear Earth–Moon dynamical environment while keeping algorithmic studies tractable. The NASA/JPL Three-Body Periodic Orbits API provides normalized periodic-orbit states, periods, Jacobi constants, stability indices, and system constants for families such as halo orbits and distant retrograde orbits [13]. The catalog-derived benchmark in this paper uses those states as transparent boundary-condition seeds. Its purpose is to expose initialization and recovery failure modes, not to certify an operational transfer.

2.2. Robust and missed-thrust trajectory design

Missed-thrust design has been studied through margin analysis, expected-thrust formulations, virtual recovery trajectories, robust optimal control, and branching trajectory methods. Rubinsztejn et al. embed expected thrust fraction into deterministic trajectory design and report improved resilience to missed-thrust scenarios [14]. Venigalla et al. formulate multi-objective low-thrust trajectory optimization with missed-thrust recovery margin [15]. Grebow and McCarty specifically study low-thrust cislunar transfers, including NRHO-to-DRO transfers, and optimize branching trajectories for outage recovery [1]. Sinha and Beeson focus on initial-guess generation for robust low-thrust missed-thrust design and compare conditional and nonconditional global-search strategies [2]. Robust cislunar low-thrust optimization has also advanced through sequential convex programming and covariance steering; Kumagai and Oguri use chance-constrained covariance steering to design nominal trajectories and correction policies under uncertainty in the Earth–Moon CR3BP [16]. Broader uncertainty-aware low-thrust design includes nonlinear programming and forward-backward shooting under uncertainty

[17], and missed-thrust robustness has recently been addressed with sequential convex programming [18]. State-dependent trust-region ideas for autonomous spacecraft SCP further illustrate the level of numerical machinery expected in robust flight-dynamics workflows [19]. These works set a high bar for any initializer: a useful discrete schedule must ultimately feed a continuous optimizer that can produce robust feasible trajectories.

2.3. QUBO/QAOA initializers

Quantum annealers and QAOA target discrete optimization models such as Ising Hamiltonians and QUBOs. A QUBO minimizes

$$E(x) = c + \sum_i a_i x_i + \sum_{i < j} b_{ij} x_i x_j, \quad x_i \in \{0, 1\}, \quad (1)$$

which is convertible to the Ising form used in quantum annealing and many gate-model workflows [20, 21]. Farhi et al. introduced QAOA as a gate-model variational algorithm for approximate combinatorial optimization [22]. Shukla and Vedula formulated trajectory optimization as a discrete search problem and argued that quantum search can provide speedups for suitably discretized cases [23]. De Grossi et al. provide the closest low-thrust QUBO precedent [3]. This paper uses QUBO and QAOA only for the discrete initialization layer; the continuous trajectory remains a classical optimal-control problem.

2.4. Benchmark gap

The targeted catalog benchmark in this paper is deliberately harder than the teacher-controlled case because cislunar transfers between stable or nearly stable periodic orbits are known to be sensitive to the continuous initial guess. The gap addressed here is therefore diagnostic rather than algorithmic replacement: the paper asks how binary schedule samplers, including QUBO/QAOA variants, behave under equal trajectory-evaluation budgets when missed-thrust penalties are included and when their schedules are handed to research-grade continuous recovery backends. Production SCP, collocation, and indirect tools remain the appropriate downstream solvers for mission-quality design. The all-windows continuous row and the continuous-backend diagnostics are the relevant controls for this benchmark layer because they test whether schedule selection improves the initial condition passed downstream. The resulting artifact package complements production-oriented studies by making failure modes, claim boundaries, and reproducibility provenance explicit.

3. Problem Formulation

3.1. Forced CR3BP dynamics

The spacecraft state is

$$s = [x, y, z, \dot{x}, \dot{y}, \dot{z}]^\top \quad (2)$$

in the normalized Earth–Moon rotating frame. With mass parameter μ , distances to the primary and secondary bodies are

$$r_1 = \sqrt{(x + \mu)^2 + y^2 + z^2}, \quad (3)$$

$$r_2 = \sqrt{(x - 1 + \mu)^2 + y^2 + z^2}. \quad (4)$$

The forced CR3BP equations are

$$\ddot{x} = 2\dot{y} + x - \frac{(1 - \mu)(x + \mu)}{r_1^3} - \frac{\mu(x - 1 + \mu)}{r_2^3} + a_x, \quad (5)$$

$$\ddot{y} = -2\dot{x} + y - \frac{(1 - \mu)y}{r_1^3} - \frac{\mu y}{r_2^3} + a_y, \quad (6)$$

$$\ddot{z} = -\frac{(1 - \mu)z}{r_1^3} - \frac{\mu z}{r_2^3} + a_z. \quad (7)$$

The transfer duration T is divided into N uniform segments. A binary variable $x_i \in \{0, 1\}$ gates thrust availability during segment i . During the discrete search stage, a deterministic target-seeking steering law maps the binary schedule to a propagated trajectory,

$$g(s) = K_r(r_f - r) + K_v(v_f - v), \quad a_i(s) = x_i a_{\max} \frac{g(s)}{\|g(s)\| + \epsilon}. \quad (8)$$

This steering law is not claimed to be optimal. It creates a repeatable map from a bitstring to a trajectory so that discrete schedules can be ranked before continuous refinement. A separate oracle diagnostic, introduced only for the teacher-controlled benchmark, initializes refinement from the hidden continuous controls used to generate the target. That diagnostic is not a competing initializer; it tests whether the refinement formulation can recover the known feasible trajectory when continuous-control initialization is no longer the bottleneck.

3.2. Missed-thrust masks and robust binary cost

Let $\mathcal{M}$ denote a set of outage masks. Each mask zeros thrust in a one- or two-segment contiguous window. For a nominal schedule x , the outage-realized schedule is $x \odot m$. The scaled terminal error under mask m is

$$e_m(x) = \left\| \begin{bmatrix} (r_T(x \odot m) - r_f)/\ell_r \\ (v_T(x \odot m) - v_f)/\ell_v \end{bmatrix} \right\|_2. \quad (9)$$

The trajectory-evaluated robust binary objective is

$$\begin{aligned} J(x) = & w_n e_0(x) + w_w \max_{m \in \mathcal{M}} e_m(x) + w_d \left(\max_{m \in \mathcal{M}} e_m(x) - e_0(x) \right) \\ & + w_p \frac{1}{N} \sum_{i=1}^N x_i + w_s \frac{1}{N-1} \sum_{i=1}^{N-1} |x_{i+1} - x_i|. \end{aligned} \quad (10)$$

The terms penalize nominal miss distance, worst outage miss distance, missed-thrust degradation, active thrust fraction, and thrust/coast switching.

4. Binary-Schedule Initializer Families

4.1. QUBO surrogate

Because $J(x)$ depends on nonlinear propagation and outage evaluation, it is not itself a QUBO. The method therefore fits

$$\hat{J}(x) = c + \sum_i a_i x_i + \sum_{i < j} b_{ij} x_i x_j \quad (11)$$

from trajectory-evaluated samples $\{x^{(k)}, J(x^{(k)})\}_{k=1}^K$ using ridge-regularized least squares:

$$\min_{\theta} \sum_{k=1}^K \left(J(x^{(k)}) - \hat{J}(x^{(k)}; \theta) \right)^2 + \lambda \|\theta\|_2^2. \quad (12)$$

The fitted QUBO is then sampled by classical QUBO simulated annealing or converted to a diagonal cost Hamiltonian for statevector QAOA. The present experiments use QAOA only as a small simulated gate-model sampler; no hardware quantum advantage is implied.

4.2. Discrete initializers

The experiments compare random search, cross-entropy search, a genetic algorithm, true-objective simulated annealing, surrogate QUBO simulated annealing, and statevector QAOA. Random search and true-objective simulated annealing evaluate propagated schedules directly. Cross-entropy updates Bernoulli segment probabilities from elite samples. The genetic algorithm uses elitism, crossover, and mutation. The QUBO and QAOA methods use true trajectory evaluations to train $\hat{J}$, then validate their best sampled schedules with the true objective.

Budget accounting is therefore reported in two ways: solver true evaluations exclude shared QUBO training, while total true evaluations include QUBO training. Equal total-evaluation comparisons are the relevant fair comparison.

4.3. Continuous branch-recovery refinement

The refinement stage converts a selected binary schedule into a continuous low-thrust candidate by optimizing piecewise-constant acceleration vectors only in active windows. Refined accelerations are projected to the Euclidean thrust ball $\|u_i\|_2 \leq a_{\max}$ before propagation and persistence. The branch-recovery mode optimizes a nominal control sequence plus selected-outage-specific post-outage recovery controls. Its least-squares residual includes nominal terminal error, selected outage terminal error, fuel regularization, control regularization, and inter-segment smoothness. A run is counted as successful only if both the nominal terminal error and the selected missed-thrust worst error are below the configured thresholds.

The paper also reports an all-mask diagnostic: after refinement on selected outages, every one- and two-segment outage mask is evaluated without claiming that all masks were optimized. This separates selected-branch recovery success from universal outage robustness.

5. Experimental Setup

5.1. Common dynamics and source state

All experiments use normalized Earth–Moon CR3BP dynamics with $\mu = 0.01215058560962404$. The common initial state is taken from the first row of an Earth–Moon L2 southern halo query with period in $[1.55, 1.65]$ TU from the NASA/JPL periodic-orbit API:

$$s_0 = [1.0324985738, 0, -0.1884844917, 0, -0.1249781287, 0]. \quad (13)$$

This state is NRHO-like in the sense that it lies in the L2 southern halo family, but the paper treats it as a normalized benchmark state, not a certified operational NRHO.

5.2. Dimensional scale and threshold interpretation

The normalized Earth–Moon scale is used only to interpret orders of magnitude. Taking the public NASA average Earth–Moon distance as 1 LU $\approx 384,400$ km [24], the recorded CR3BP normalization gives 1 TU $\approx 375,190.259$ s = 4.34248 days, velocity unit ≈ 1.02455 km/s, and acceleration unit ≈ 2.73074 mm/s². Thus the main phase-shift transfer time $T = 0.5$ TU is about 2.17 days, while the hard-catalog tail-coast recovery case with $T = 4.0$ TU is about 17.37 days. The acceleration bounds used in the phase-shift and high-thrust diagnostics correspond to $a_{\max} = 0.2 \approx 0.546$ mm/s² and $a_{\max} = 1.0 \approx 2.731$ mm/s², respectively.

The terminal-error thresholds are screening tolerances in the normalized metric

$$\sqrt{\|\Delta r\|_2^2 + \|\Delta v\|_2^2/0.35^2}, \quad (14)$$

not flight navigation or operational targeting tolerances. If velocity error is zero, the nominal threshold 0.09 corresponds to a position-only error of about 34,596 km, and the selected-recovery threshold 0.17 corresponds to about 65,348 km. If position error is zero, the same thresholds correspond to velocity-only errors of about 32.3 m/s and 61.0 m/s under the configured velocity scale. This scale is the reason the manuscript frames the results as preliminary design and initializer benchmarking rather than operational targeting.

5.3. Benchmark families

Two benchmark families define the main evidence path. The harder catalog-derived family uses a distant retrograde orbit seed from an Earth–Moon DRO query,

$$s_{\text{DRO}} = [0.8161260827, 0, 0, 0, 0.5076556931, 0], \quad (15)$$

and generates the target by ballistic propagation of that seed. The original catalog-derived schedule run uses $T = 2.7$ TU, $N = 12$, $a_{\max} = 0.055$, and one- and two-segment outage masks. It remains negative evidence in the supplement. The main hard-catalog result instead reports a later locked-nominal tail-coast continuous-backend diagnostic at $T = 4.0$ TU, $N = 14$,

and $a_{\max} = 1.0$, where the final five nominal control segments are constrained to zero and all configured one- and two-segment masks are selected and evaluated in one fixed-final-time row.

The non-teacher phase-shift family is derived from the same JPL L2 southern halo source state s_0 . The target is generated by zero-thrust CR3BP propagation of s_0 for a phase time of 0.3 TU, while the transfer must arrive at that target after $T = 0.5$ TU. A zero-thrust trajectory has normalized terminal error 0.3758, so the case is not a do-nothing pass. The primary initializer benchmark uses $N = 12$, $a_{\max} = 0.2$, one-segment outage masks, and a duty-cycle prior

$$J_\rho(x) = w_\rho \left(\frac{1}{N} \sum_{i=1}^N x_i - \rho \right)^2, \quad (16)$$

with $\rho = 11/12$ and $w_\rho = 12$. This prior is not an oracle trajectory correction; it encodes a preference for one coast window while preserving a dense low-thrust arc.

The supplement preserves the secondary diagnostic families: the catalog pilot and QUBO fit diagnostics, feasibility and direct multiple-shooting stress tests, locked-nominal and delayed-arrival hard-catalog recovery packages, bounded phase-suite and robust-margin rows, direct-collocation and Hermite–Simpson baselines, dense-repair and legacy cardinality ablations, and the teacher-controlled feasible benchmark. These diagnostics explain provenance and failure modes but are not promoted to headline evidence in the main article.

5.4. Metrics

The primary metrics are refined nominal terminal error, refined selected missed-thrust worst error, all-mask diagnostic worst error, schedule active fraction, nominal and recovery fuel, total true trajectory evaluations, and refinement success rate. A run is counted as a robust refinement success only when both the nominal terminal error and the selected missed-thrust worst error are below the configured thresholds. Unless otherwise stated, those thresholds are 0.09 for nominal success and 0.17 for selected robust recovery success.

The main $N = 12$ duty-cycle-prior statistics package repeats seven methods over 30 deterministic seeds, for 210 raw method–seed rows,

and reports Wilson success intervals, paired per-seed success differences, bootstrap confidence intervals for mean selected-worst-error differences, and exact two-sided sign tests. A separate focused QAOA-depth package repeats the QUBO-derived methods over 30 deterministic seeds, for 150 raw method-seed rows. The hard-catalog tail-coast row `tail_coast_all_one_two_segment_t5_portfolio` uses the same thresholds, selects all configured one- and two-segment masks in one case row, and reports optimizer convergence separately from threshold feasibility so late-tail direct evaluations without recovery variables are not mistaken for optimizer-converged branch solves.

6. Results

The main results are screening evidence for a normalized CR3BP benchmark. The dimensional caveat in Section 5.2 applies to every table: the 0.09/0.17 thresholds correspond to large position-only or velocity-only errors and should not be read as operational targeting tolerances. Secondary tables and figures are retained in the supplement so that the main article follows the primary claim path, while the evidence-synthesis table below cross-indexes representative rows from those recorded artifacts.

6.1. Claim hierarchy

Table 1 records the claim boundary used throughout the paper. It separates the benchmark contribution, the equal-budget initializer comparison, continuous-backend diagnostics, and claims that remain outside the evidence.

6.2. Reviewer-facing evidence ledger

Table 2 is generated by `scripts/run_claim_evidence_ledger.py`. The script reads only recorded artifacts and writes `data/results/claim_evidence_ledger/claim_evidence_ledger.csv`, `data/results/claim_evidence_ledger/claim_evidence_ledger_metadata.json`, and the LaTeX tables used here. It is not a trajectory-optimization run. Its revision purpose is narrower than the cross-backend synthesis: it separates selected-branch evidence, all-mask diagnostics, and rows where every configured mask was actually selected and evaluated.

The ledger makes the claim boundaries auditable. The 30-seed initializer and QAOA/QUBO rows are selected-branch statistical evidence and preserve the negative sampler conclusion. The continuation and hard-catalog portfolio

rows are all-configured-mask evidence only within their stated mask families. The accepted-control replay row is a deterministic normalized-CR3BP replay of persisted nominal and branch controls, not a new optimizer run. The direct-collocation and independent-midpoint Hermite–Simpson rows retain the selected-branch/all-mask distinction rather than promoting all-mask diagnostics to robustness claims.

For practitioners, the reporting lesson is concrete: include the all-windows continuous control, do not collapse selected-branch and all-mask semantics, preserve negative QAOA/QUBO evidence, and archive provenance in machine-readable artifacts. Tables 3 and 4 add two recorded-error tail-coast checks for the historical four-row package. The combined hard-catalog tail-coast row passes recorded-error audits down to $(0.025, 0.095)$, fails when the robust threshold is tightened to 0.09, and also fails the very-tight 0.02 nominal threshold. The branch audit records 27 total branches, 25/25 optimizer-converged eligible branches, two no-recovery-variable direct evaluations, and four accepted fallback starts. The focused accepted-control replay package is now present, so the ledger includes a separate normalized-CR3BP replay row for the persisted nominal control and all 27 accepted branch-control sidecars.

6.3. Cross-backend evidence synthesis

Table 5 is generated by `scripts/run_evidence_synthesis.py`. The script reads only recorded CSV/JSON artifacts and writes `data/results/evidence_synthesis/evidence_synthesis.csv`, `data/results/evidence_synthesis/evidence_synthesis_metadata.json`, and the LaTeX table used here. It is not a new optimization run. Its purpose is to put the broader validation axes into the main claim path without overstating them.

The synthesis clarifies the positive contribution. The 30-seed threshold-sensitivity row shows that at the tight $(0.050, 0.090)$ check every sampled method is 0/30 while all-windows continuous remains 30/30, so the paper does not claim sampler superiority. The continuation-extension rows show that all-mask direct multiple-shooting continuation can meet a tighter $(0.065, 0.100)$ phase-shift screen for all one-segment masks and for all configured one/two masks, but not the 0.05 nominal headroom check. The direct-collocation and independent-midpoint Hermite–Simpson phase-shift rows show compact continuous backends with tight selected/all diagnostics, while preserving the distinction between optimized selected branches and all-mask diagnostics. The hard-catalog rows show that the fixed-final-time

pass is scoped to the tail-coast locked-nominal backend; an independent-HS selected-outage hard-catalog diagnostic remains a negative result.

6.4. Recorded-control replay/stress validation

Table 6 reports a deterministic replay over persisted nominal-control sidecars, generated by `scripts/run_replay_stress_validation.py`. The script reads the representative continuation-extension rows `all_single_p04_warm_from_p03` and `two_segment_n8_p03_cold`, plus the independent-midpoint Hermite–Simpson row `ihp_phase_p04_amax02_warm_from_p03`. It then repropagates only the recorded nominal controls under the source substep count, two refined integration grids, and direct $\pm 1\%$ acceleration-scale perturbations. No least-squares optimization, schedule search, branch recovery optimization, or high-fidelity force model is launched.

The source-grid baseline reproduces the recorded nominal errors to within 7.7×10^{-17} across the three rows. Refining the integration grid changes the nominal errors by at most about 1.5×10^{-8} . The direct acceleration-scale stress rows remain below the configured nominal screening threshold for all three cases; only the independent-midpoint Hermite–Simpson case remains below the 0.05 nominal check. Scale-up rows may exceed the configured acceleration bound and are reported only as stress diagnostics. This table therefore supports sidecar determinism and nominal replay stability, not high-fidelity validation, operational targeting, or missed-thrust branch robustness.

6.5. Tail-coast hard-catalog diagnostic

Table 7 reports the strongest hard-catalog fixed-final-time result. The setup solves and freezes a locked nominal control, then refines it with the final five nominal control segments constrained to exact zero. Branches keep the original target and original final time; no delayed target or added recovery horizon is used. The nominal-only provenance row has tail-coast nominal error 0.022992 and therefore passes the nominal screening threshold, while its masked-nominal all-mask diagnostic remains 27.8351 because no branch recovery is run.

The headline row is `tail_coast_all_one_two_segment_t5_portfolio`. It sets `outage_lengths=[1,2]`, selects all configured masks, and evaluates all 27 selected masks in one fixed-final-time case row. The row reaches tail-coast nominal error 0.02299233817855882, selected fixed-time worst

error 0.0936063931709301, and all-mask fixed-time diagnostic worst error 0.0936063931709301, so it meets the configured 0.09/0.17 screening thresholds. The branch portfolio evaluates deterministic fallback starts for four branches, charges 5929 function evaluations, and records runtime about 1833.6 s in the current CSV. Among branches with post-outage recovery variables, the optimizer-converged count is 25/25. The remaining two late-tail masks have no post-outage recovery variables; they are direct fixed-final-time threshold evaluations rather than optimizer-converged branch solves. Therefore `branch_optimizer_all_success=false` remains a conservative CSV accounting flag even though `meets_thresholds=true`.

This result is deliberately scoped. It is fixed-final-time continuous-backend evidence with a locked nominal control, final-five-segment coast, configured one- and two-segment masks, fallback starts, and late-tail no-recovery-variable accounting. It is not a binary-sampler result, not QUBO/QAOA evidence, not fuel optimality, not high-fidelity validation, and not robustness to outage families beyond the configured masks.

6.6. Accepted branch-control replay

Table 8 reports the completed focused accepted branch-control replay for the same combined hard-catalog tail-coast case. The recovery package persists a nominal-control sidecar, one accepted full-control sidecar per selected branch, a progress CSV, and a manifest with SHA-256 hashes, target state, thresholds, settings fingerprint, and explicit limitations. The replay post-processor reads those recorded artifacts and repropagates them under the configured normalized CR3BP model.

The replay CSV contains one nominal row and 27 branch rows. The generated metadata reports zero nominal terminal-error delta, zero maximum branch terminal-error delta, zero maximum terminal-error delta, tolerance 10^{-10} , and `passes_tolerance=true`. This supports deterministic accepted-control replay for the persisted controls only. It does not rerun the nominal solve, branch optimizer, portfolio selection, fallback search, schedule search, or any high-fidelity force model, and it is not production solver parity, fuel optimality, QUBO/QAOA, or quantum-advantage evidence.

6.7. Thirty-seed main-method statistics

Tables 9 and 10, together with Figure 1, repeat the configured $N = 12$ duty-cycle-prior method comparison over 30 deterministic seeds. The raw package contains 210 rows, i.e., 30 seeds times seven methods: random

search, cross-entropy search, genetic search, true-objective simulated annealing, surrogate-QUBO simulated annealing, the configured statevector QAOA sampler, and all-windows continuous refinement.

The success-rate evidence strengthens the 10-seed result but does not create a quantum-performance claim. Random search succeeds in 11/30 seeds, with Wilson 95% interval $[0.219, 0.545]$. Cross-entropy, genetic search, true-objective simulated annealing, surrogate-QUBO simulated annealing, and all-windows continuous refinement each succeed in 30/30 seeds, with Wilson interval $[0.886, 1.000]$. The configured `qaoa_statevector` method succeeds in 13/30 seeds, with interval $[0.274, 0.608]$.

The paired error comparisons are more important for the aerospace interpretation. Against all-windows continuous refinement, every sampled method has a positive mean selected-worst-error difference; negative would favor the sampled method. The exact two-sided sign-test value is approximately 1.86×10^{-9} for each sampler versus all-windows continuous refinement. Thus, the all-windows continuous baseline is not just equally successful for the strongest methods; it also gives lower selected-worst error on this benchmark. Against surrogate-QUBO simulated annealing, cross-entropy, genetic search, and true-objective simulated annealing tie the success rate and have small nonsignificant selected-worst-error deltas. Random search is worse. The configured statevector QAOA row has lower success and a positive mean selected-worst-error delta of 0.0153 versus surrogate-QUBO simulated annealing, with exact sign-test $p = 0.1849$.

This package therefore strengthens the statistics beyond a QAOA-only ablation while preserving the conservative conclusion. Duty-cycle priors help several samplers find feasible high-duty schedules, but all-windows continuous fuel regularization remains a stronger mechanism for selected-worst error in this benchmark, and the evidence does not support quantum advantage or QAOA superiority.

Table 11 adds a threshold-sensitivity sanity check computed only from `data/results/phase_shift_cardinality_30seed/raw_results.csv`. At the stricter $(0.075, 0.120)$ nominal/selected-worst thresholds, cross-entropy, genetic search, true-objective simulated annealing, surrogate-QUBO simulated annealing, and all-windows continuous refinement still pass 30/30 seeds, while random search falls to 4/30 and `qaoa_statevector` remains 13/30. At $(0.065, 0.100)$ the sampled high-duty schedules largely stop passing, but all-windows continuous remains 30/30. At the tight $(0.050, 0.090)$ check, every sampled method is 0/30 while all-windows continuous is still 30/30. Thus

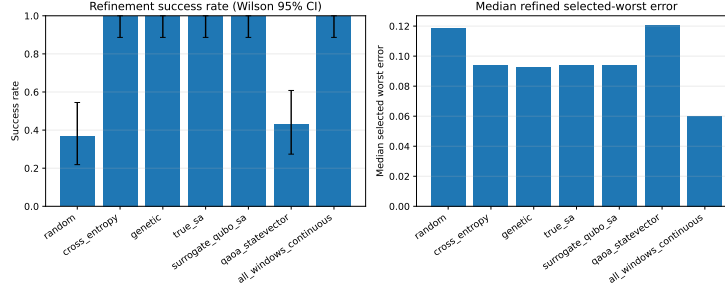


Figure 1: Thirty-seed main-method statistics for the duty-cycle-prior benchmark. High-duty samplers can be feasible, but all-windows continuous refinement remains lower-error on paired selected-worst comparisons.

stricter thresholds do not rescue sampled-method superiority; they quantify the margin limitation of the sampled schedules and reinforce the all-windows continuous lower-error diagnostic.

6.8. Focused QAOA/QUBO ablation

Table 12 and Figure 2 isolate the QUBO-sampling step on the same $N = 12$ duty-cycle-prior benchmark with 30 deterministic seeds. Each QUBO-derived method uses the same accounting per seed: 96 shared trajectory evaluations to train the QUBO and 24 true post-QUBO candidate evaluations. The resulting raw package contains 150 method–seed rows; the raw method summary table is in the supplement.

The 30-seed success rates with Wilson 95% intervals are $28/30 = 0.933$ $[0.787, 0.982]$ for surrogate-QUBO simulated annealing, $28/30 = 0.933$ $[0.787, 0.982]$ for exact evaluation of the top 24 QUBO bitstrings, $15/30 = 0.500$ $[0.332, 0.668]$ for random-angle $p = 1$ QAOA, $21/30 = 0.700$ $[0.521, 0.833]$ for optimized $p = 1$ QAOA, and $29/30 = 0.967$ $[0.833, 0.994]$ for optimized $p = 2$ QAOA. Optimized $p = 2$ is competitive with the strongest classical QUBO sampler in this narrow statevector QUBO ablation, but it is not statistically supported as superior: versus surrogate-QUBO simulated annealing, its paired success delta is only $+0.033$, the mean selected-worst-error delta is -0.00056 with bootstrap 95% confidence interval $[-0.00393, 0.00236]$, and the exact sign-test p -value is 0.4545.

The useful positive result is narrower. Relative to random-angle $p = 1$ QAOA, optimized $p = 2$ QAOA increases success by $+0.467$ and reduces the mean selected-worst error by 0.0125 with bootstrap interval

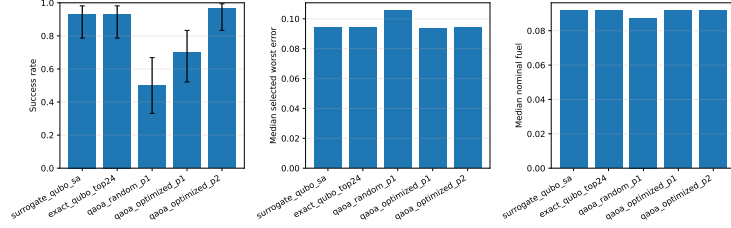


Figure 2: Thirty-seed QAOA-depth ablation summary. Optimized $p = 2$ QAOA improves over random-angle QAOA and is competitive with classical QUBO sampling, but paired tests do not support a superiority claim over surrogate-QUBO simulated annealing.

$[-0.0188, -0.0060]$. The exact sign-test p -value is still not significant at conventional levels ($p = 0.2478$), so the evidence should be read as practical improvement in this ablation rather than a formal dominance claim. No quantum advantage is claimed.

6.9. Supplementary diagnostics

The supplement preserves the evidence that explains how the headline results were reached and where they fail. This includes the original catalog pilot and QUBO fit, feasibility sweeps, direct multiple-shooting and targeted catalog attempts, selected-outage hard-catalog envelope, locked-nominal and delayed-arrival hard-catalog recovery packages, bounded phase-suite and robust-margin rows, direct-collocation and Hermite–Simpson baselines, unrepaired and repaired $N = 8$ phase-shift evidence, legacy 10-seed cardinality-prior summaries, cardinality and fuel ablations, and the teacher-controlled benchmark. These artifacts support the failure-mode discussion, but the main claims do not depend on promoting them to primary results.

7. Discussion

The central result is a benchmark result, not a sampler-performance claim. Binary thrust-window schedules can be embedded in a reproducible missed-thrust initialization workflow, but the usefulness of a schedule is controlled by the continuous recovery basin. The supplementary diagnostics show both sides: sparse binary schedules can fail even when all-windows continuous refinement succeeds, while duty-cycle priors can preserve enough thrust availability for several samplers to find refinable high-duty schedules.

The 30-seed main-method package is the most relevant initializer evidence. Cross-entropy search, genetic search, true-objective simulated annealing, surrogate-QUBO simulated annealing, and all-windows continuous refinement all succeed in 30/30 seeds on the duty-cycle-prior benchmark, whereas the configured statevector QAOA row succeeds in 13/30 seeds. The paired selected-worst-error tests favor all-windows continuous refinement over every sampled method, and the stricter-threshold sensitivity check shows the sampled high-duty schedules losing pass counts before all-windows continuous does. Thus the paper supports duty-cycle-aware robust initialization as a useful benchmark mechanism, but not a claim that a binary sampler improves the continuous trajectory beyond a strong dense-control baseline.

The focused QAOA/QUBO ablation provides a more favorable but still bounded view of QAOA. Optimizing the angles and increasing simulated depth to $p = 2$ improves QAOA relative to random-angle $p = 1$ QAOA and is competitive with classical QUBO sampling in that narrow statevector study. The paired comparisons do not support superiority over surrogate-QUBO simulated annealing, and the result is not hardware evidence. For this benchmark, QAOA is best interpreted as a measured initializer family with negative and inconclusive comparisons, not as the contribution itself.

The hard-catalog tail-coast diagnostic addresses a different reviewer concern: whether any hard-catalog missed-thrust recovery case can meet the configured fixed-final-time thresholds once nominal feasibility is locked. The combined one/two-segment row does meet those thresholds, but only within a carefully scoped continuous-backend setup. The final-five-segment coast, locked nominal control, fallback starts, 25/25 optimizer-converged eligible recovery branches, and two no-recovery-variable late-tail direct evaluations are part of the result. This narrows a backend blocker and records useful recovery provenance, but it does not establish fuel-optimality, high-fidelity robustness, broader outage-family robustness, or quantum evidence.

The practical implication for astrodynamics researchers is that binary schedule benchmarks should report continuous-backend diagnostics and negative evidence as first-class artifacts. A stronger future initializer claim would need better trajectory-shaped binary encodings, mature continuous optimal-control baselines such as production collocation or sequential convex programming workflows, high-fidelity validation, and statistically significant improvements over strong classical initializers under equal total trajectory-evaluation budgets.

8. Limitations

The normalized CR3BP experiments are preliminary-design algorithm studies. They omit ephemeris dynamics, mass depletion, eclipse constraints, navigation uncertainty, thrust pointing limits beyond the acceleration norm, operational constraints, and high-fidelity force modeling. The terminal-error thresholds are screening tolerances; their dimensional interpretation is deliberately large and not comparable to flight targeting requirements.

The continuous backends are research-grade direct shooting, compact direct collocation, Hermite–Simpson, and continuation diagnostics. They are useful for comparing initializer behavior and exposing failure modes, but they are not certified flight-design tools. The recorded-control replay/stress table checks deterministic nominal sidecar replay only. The focused tail-coast accepted-control replay checks persisted branch-control sidecars under normalized CR3BP only; it is not an optimization rerun, high-fidelity validation, production solver parity, fuel-optimal analysis, or quantum evidence. The claim evidence ledger clarifies selected-branch, all-mask diagnostic, all-configured-mask, and accepted-control replay semantics, but it does not add high-fidelity validation. Many supplementary rows optimize selected outage branches and report unselected outage masks diagnostically; only rows that explicitly select all configured masks should be read as all-configured-mask evidence.

The hard-catalog tail-coast result is scoped to a locked-nominal continuous-backend setup with final-five-segment coast, configured one- and two-segment masks, deterministic fallback starts, and late-tail no-recovery-variable direct evaluations. It does not prove fuel-optimal recovery, high-fidelity robustness, certified flight design, or broader outage-family robustness.

The discrete-initializer conclusions are also bounded. The 30-seed packages improve statistical rigor, but QAOA is simulated with shallow depths and limited optimizer budgets, and the evidence does not support quantum advantage or QAOA superiority. Smaller supplementary packages, including teacher-controlled and legacy 10-seed results, are retained as diagnostics rather than as strong stochastic rankings.

9. Reproducibility

The repository contains Python source, configurations, source-state metadata, generated tables, figures, result metadata, and PDFs. The

primary artifacts for the main claims are the claim-evidence ledger in `data/results/claim_evidence_ledger/`, the evidence-synthesis package in `data/results/evidence_synthesis/`, the replay/stress sidecar package in `data/results/replay_stress_validation/`, the 30-seed main-method package in `data/results/phase_shift_cardinality_30seed/`, including the raw-CSV-only threshold-sensitivity artifacts, the 30-seed QAOA/QUBO package in `data/results/qaoa_depth_ablation_30seed/`, the hard-catalog tail-coast package in `data/results/hard_catalog_tail_coast_recovery/`, and the focused hard-catalog accepted branch-control replay package in `data/results/hard_catalog_tail_coast_branch_control_replay/`. The supplement and `REPRODUCIBILITY.md` map the secondary diagnostics to their tables, figures, commands, and known caveats.

The reviewer-facing verification path is: start from a clean worktree, re-run the deterministic ledger, evidence-synthesis, replay/stress, and accepted branch-control replay postprocessors, rebuild `paper/main.pdf` and `paper/supplement.pdf` with `latexmk`, run the smoke tests with Python 3.11, refresh `data/results/artifact_manifest.json`, run the manifest checker, and run `git diff -check`. Long experiment jobs are not part of the short verification path; recorded artifacts should be used unless evidence is intentionally regenerated.

The manifest is generated by `scripts/write_artifact_manifest.py`. Its scoped SHA-256 hashes and byte counts are the authoritative artifact identities. The field `git_head_at_generation` records the repository state when the manifest was produced; for a committed manifest this necessarily refers to the parent/pre-final state rather than the final commit containing the refreshed manifest. Historical long-run metadata may predate the repository or contain dirty-state records. The current submission snapshot is verified by clean git status after final commit, manifest checking, tests, and local LaTeX builds rather than by interpreting old experiment metadata as final provenance.

10. Conclusion

This paper contributes a reproducible normalized CR3BP benchmark for robust low-thrust cislunar initialization under missed-thrust events. Its main finding is conservative: duty-cycle-aware binary schedules can help locate refinable high-duty candidates, but all-windows continuous refinement remains lower-error in paired selected-worst comparisons and remains more

robust under stricter threshold checks, while the tested simulated QAOA variants do not establish superiority over classical initializers. Deterministic evidence-ledger, evidence-synthesis, and sidecar-replay layers make the broader validation axes explicit: all-mask continuation and compact collocation can provide phase-shift headroom, recorded nominal controls replay consistently, the hard-catalog pass still requires the scoped tail-coast backend, and the accepted hard-catalog branch-control sidecars replay exactly under the configured normalized CR3BP model. The useful outcome is therefore an auditable benchmark with negative evidence, failure modes, and provenance, not a claim of quantum advantage or flight-ready robustness.

Declaration of Competing Interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability

All code and generated artifacts are available in the working repository associated with this draft. The benchmark source states are derived from the NASA/JPL Three-Body Periodic Orbits API and recorded in `data/source_states.json`.

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Table 1: Manuscript-level claim-to-artifact map. The table separates primary claims, diagnostic evidence, and claims outside the evidence boundary.

Claim layer	Artifact map	Defensible interpretation	Explicit boundary
Binary-schedule benchmark	Source states, configs, result CSV/JSON, generated tables/figures, and artifact manifest	The paper provides an auditable normalized CR3BP missed-thrust initialization benchmark	Preliminary-design screening only; not operational targeting or flight certification
Equal-budget initializer comparison	Duty-cycle-prior 30-seed raw results, success intervals, paired comparisons, threshold sensitivity, and tables	Several classical and surrogate-QUBO samplers find refinable high-duty schedules at the configured threshold, but stricter thresholds expose limited sampled-method margin	No quantum advantage or QAOA superiority; all-windows continuous refinement remains lower-error on paired selected-worst comparisons and remains 30/30 under the stricter checks
Continuous-backend diagnostics	Bounded multiple shooting, continuation, direct collocation, Hermite-Simpson, replay/stress sidecar validation, and sidecar metadata packages	Several non-teacher normalized CR3BP phase-shift rows are feasible with norm-bounded continuous controls, and their persisted nominal controls replay deterministically	Research-grade shooting, collocation, and SCP-adjacent diagnostics only; not certified flight-design tools or high-fidelity validation
Hard-catalog tail-coast recovery and replay	Tail-coast recovery and focused replay CSV/table/metadata rows	The combined row selects/evaluates all 27 configured one- and two-segment masks, and the persisted accepted controls replay deterministically under normalized CR3BP	No quantum, fuel-optimal, high-fidelity, production-parity, or broader outage-family claim
Open claims and failure modes	Negative hard-catalog envelopes, limitations, metadata, and supplementary diagnostics	The package records unresolved cases and failure modes rather than promoting every diagnostic to a headline result	Requires stronger continuous-control initialization, broader validation, high-fidelity dynamics, and statistically significant improvement over strong classical initializers

Table 2: Reviewer-facing claim evidence ledger generated from recorded artifacts only. The table is intentionally semantic: it distinguishes selected-branch evidence, all-mask diagnostics, all-configured-mask evidence, and accepted-control replay without adding new optimization results.

Case	Configuration	Test	Pass/Fail	Reason	Pass/Fail	Reason
tail_coast_all_one_two_segment_t5_portfolio	tail_coast_all_one_two_segment_t5_portfolio	tail_coast_all_one_two_segment_t5_portfolio	True	tail_coast_all_one_two_segment_t5_portfolio	True	tail_coast_all_one_two_segment_t5_portfolio

Table 3: Tail-coast recorded-error threshold audit for tail_coast_all_one_two_segment_t5_portfolio. The audit uses only recorded nominal, selected-worst, and all-mask-worst errors; no optimization is rerun.

Threshold pair	Nominal threshold	Selected/all threshold	Nominal pass	Selected pass	All-mask pass	Pair pass
configured_0p09_0p17	0.09	0.17	True	True	True	True
screen_0p05_0p12	0.05	0.12	True	True	True	True
near_margin_0p025_0p095	0.025	0.095	True	True	True	True
robust_tight_0p025_0p09	0.025	0.09	True	False	False	False
very_tight_0p02_0p09	0.02	0.09	False	False	False	False

Table 4: Tail-coast branch audit parsed from the historical **branch_results** JSON field. Counts summarize recorded branch outcomes only; the accepted-control sidecar replay is a separate focused package.

Case	Branches	Optimizer run	Optimizer success	Direct no-recovery	Failback accepted	Max terminal error	Accepted weights	Accepted init kinds
tail_coast_all_one_two_segment_t5_portfolio	27	25	25	2	4	0.09060331709301	regularized_001-6; terminal_only-21	constant_vector-4; no_recovery_variables-2; nominal_post_outage-21

Table 5: Cross-backend evidence synthesis generated from recorded artifacts only. Error entries are nominal / selected-worst / all-mask-worst in the normalized terminal metric. Tighter pass statuses use all-mask diagnostics conservatively; selected-branch rows remain diagnostics unless all configured masks are explicitly selected.

Case	Configuration	Test	Pass/Fail	Reason	Pass/Fail	Reason
tail_coast_all_one_two_segment_t5_portfolio	tail_coast_all_one_two_segment_t5_portfolio	tail_coast_all_one_two_segment_t5_portfolio	True	tail_coast_all_one_two_segment_t5_portfolio	True	tail_coast_all_one_two_segment_t5_portfolio

Table 6: Deterministic nominal-control replay/stress validation from recorded sidecars. Pass flags are nominal-error checks only because this script does not rerun selected missed-thrust branch recovery. Acceleration scale-up rows are direct stress diagnostics and may exceed the configured acceleration bound; they are not high-fidelity validation.

Case	Configuration	Test	Pass/Fail	Reason	Pass/Fail	Reason
tail_coast_all_one_two_segment_t5_portfolio	tail_coast_all_one_two_segment_t5_portfolio	tail_coast_all_one_two_segment_t5_portfolio	True	tail_coast_all_one_two_segment_t5_portfolio	True	tail_coast_all_one_two_segment_t5_portfolio

